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Attending to Body and Mind: Does Interoceptive Attention Compete With Controlled and Negative Automatic Thoughts?

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Interoceptive attention refers to how an individual typically allocates their attentional resources to physiological signals. We do not, however, fully understand how interoceptive attention relates with other forms of internally focused attention, such as attention to internal thoughts (e.g., mind-wandering, reflection, worry), and whether these factors compete for limited attentional resources at the trait and state level. Across two studies (n = 222, n = 109; years data collected: 2021–2022, 2023–2024; from a U.K.-based predominantly undergraduate population), we isolated two components from several established trait-level measures of attention to thoughts, these reflected negative automatic thoughts (worry, rumination) and controlled thoughts (deliberate mind-wandering, reflection). In subsequent regression analyses, negative automatic thoughts, but not controlled thoughts, were a significant positive predictor of interoceptive attention, even when controlling for perceived interoceptive accuracy. This significant relationship was, however, accounted for by trait anxiety when included in the model. To test this pattern at the state-level, a novel monotonous responding task with three-dimensional attention probes was developed, which measures the reported allocation of attention between thoughts, bodily sensations, and the task. In contrast to trait-level attention, reported priority of attention to thoughts and body signals was inversely correlated, despite both increasing across the task. The results suggest that in a single moment attention to bodily signals and internal thoughts may compete; but over time, individuals who report focusing more on their internal thoughts also report more time attending to their body, with some evidence showing that this positive correlation could be due to underlying trait anxiety.

Keywords: interoceptive attention, interoception, internal attention, negative automatic thoughts, mind-wandering

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Interoception is the experience and awareness of the physiological condition of the body, that allows sensing, interpreting, and integration of signals originating from within the body (Khalsa et al., 2018). The ability to detect and subsequently respond to our visceral signals is important to maintaining homeostatic systems as well as regulating our emotions, due to the physiological component underpinning our emotional states (Critchley & Garfinkel, 2017; Füstös et al., 2013).

Conventionally, interoception has been measured as the accuracy with which an individual can perceive a physiological signal, such as when counting one's own heartbeats; or the subjective ability to perceive one's own physical states, as measured through questionnaires (Garfinkel et al., 2015). More recently, however, a self-report scale has been developed which measures the amount of interoceptive attention allocated to physiological signals—rather than the perceived ability to correctly track these signals (Gabriele et al., 2022; Murphy et al., 2020). Interestingly, interoceptive attention is distinctive from objective or subjective interoceptive accuracy or sensibility, as it measures the frequency and time spent attending to physiological signals, independent of actual or perceived accuracy of this detection.

Previous research has often explored interoception in relation to biological sensation and neural integration of signals from disparate neural and afferent nerves (Craig, 2002, 2003; Herbert & Pollatos, 2012; Wiens, 2005). However, the idea of measuring interoception as a form of limited capacity attention, which can be focused on physiological signals, raises the possibility that it can covary with other forms of attention such as those to our internal streams of thought or external stimuli. Indeed, there is great variation across individuals in the extent they are disposed to different trait-like attentional states, such as mind-wandering, worry, reflection, and rumination (Gustavson et al., 2018; Seli, Risko, Smilek, & Schacter, 2016).

One possibility is that there may be a general disposition for internal or self-focused attention, which encompasses both attention

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The study materials, data, and scripts for analyses for all studies are publicly accessible via the Open Science Framework, access code at https://osf.io/wjc45/?view_only=46141f0dc32147509bae08b15e9a9a8f. The authors declare no financial or nonfinancial conflicts of interest.

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to bodily signals as well as internal thoughts. Indeed, evidence suggests that spontaneous physical sensations in the hands can be positively correlated with the degree that individuals are prone to spontaneous mind-wandering, suggesting a common self-focused internal attentional mechanism (Michael et al., 2017). Similarly, self-reported focus on one's body and physical movement (i.e., fidgeting) was positively correlated with self-reported frequency of mind-wandering (Carriere et al., 2013). Thus, individuals who are disposed to attend to physical sensations could also be more prone to attend to internal streams of thought, due to an underlying attentional trait tendency to prioritize all forms of "internal stimuli."

On the other hand, it could be possible that interoceptive attention to bodily signals is distinct from attention to streams of internal thought. In which case, due to limited attentional capacity which they must compete for, interoceptive attention and attention to thoughts would be inversely correlated. Consistent with this competitive relationship, evidence suggests that when participants are instructed to engage in interoceptive attention by focusing on their breathing, their rates of mind-wandering are reduced (Brewer et al., 2011; Levinson et al., 2014; Lutz et al., 2008; Weng et al., 2020). Further, when participants' attention is engaged with a concurrent working memory task, they report reduced bodily sensation and feeling of bodily agency, suggesting reduced attention or integration of visceral signals due to competition for limited attentional capacity (Qu et al., 2021; Salgues, Plancher, & Michael, 2021).

Based on this competitive internal attention hypothesis, it would be predicted that measures of trait attention to internal thought (i.e., worry, rumination, mind-wandering, reflection) will inversely correlate with self-reported interoceptive attention. Conversely, if interoceptive attention is positively related with reported frequency of internally focused thoughts, it likely shares a similar underpinning or may even reflect a different facet of the same construct.

The current investigation therefore aimed to clarify the relationship between these two factors. Specifically, in two samples, trait-level self-reported interoceptive attention and other traits that comprise attention to internal thoughts were correlated to assess whether these reflect a competitive or common relationship. Attention to internal thoughts, however, is not a unitary construct and varies along several dimensions including how controlled or automatic the thoughts are, their valence, and their temporal focus (i.e., future/past; Hallion et al., 2022). In order, therefore, to reduce the measures of internal thought down to their core components, we collected responses on multiple established measures of self-reported attention to internal thoughts, and condensed them using principal components analysis (PCA). These measured constructs were rumination, reflection, worry, and deliberate and spontaneous mind-wandering.

Further, it is possible that the proposed competitive relationship between internal thoughts and interoceptive attention could be underpinned by other factors. For instance, those who experience high trait anxiety often also experience high levels of negative automatic thoughts and interoceptive attention, as well as lower levels of interoceptive accuracy (Craig, 2003; Tsakiris & Critchley, 2016). We therefore included a measure of anxiety to explore whether it could underpin the proposed competitive relationship.

Additionally, both interoceptive attention and attention to thoughts have been implicated in the processing of emotions, with both forms often linked to emotional processing and regulation deficits (Marchetti et al., 2016; Murphy et al., 2018; Zamariola et al., 2019). To assess whether these different trait-level attentional mechanisms show a similar relationship with emotional processing, consistent with a related mechanism or function, we included measures of emotional clarity (i.e., perception/identification) and emotional awareness.

The trait-level exploration of the relationships was conducted in both Studies 1 and 2, though in Study 2 we also explored whether the same patterns emerged at the state-level, when participants reported their immediate focus of attention in a single moment. To do this, we developed a novel monotonous responding task with three-dimensional attention probes. Within this task, participants were required to complete a simple button-pressing task using their keyboard, and at random intervals, their attention was probed using a three-dimensional rating scale asking them how they divided their attention between (1) the task, (2) their internal thoughts, and (3) their bodily sensations. This three-dimensional scale forced the participants to identify where their attention was focused within a single moment, thus perfectly isolating the state-level allocation of attention between the different streams of sensory/cognitive information.

The present study is therefore able to explore whether both reported trait and state interoceptive attention and attention to internal thoughts reflect distinct competitive factors (i.e., inversely related factors) or whether they reflect common factors (i.e., positively related factors).

Method

Participants

Study 1

Participants were recruited via social media (i.e., WhatsApp groups and Facebook groups), and the student participant pool in exchange for course credits. An initial sample of 316 participants responded to the survey, 62 were excluded for not completing the whole survey, and a further 31 were excluded for repetitive responding. Repetitive responders were identified by recording the number of times a participants' response was identical to a previous item worded in the opposing direction on scales with reverse scored items (i.e., Penn State Worry Questionnaire and Rumination–Reflection Questionnaire). Those who scored over 2 *SD*s on this measure were excluded (procedure originally developed for Brown et al., 2023). The final sample of 222 participants consisted of 183 female and 37 male participants and one who preferred not to report their gender. The average age of participants was 23.90 ($SD = 7.97$; range = 18–70) years old.

The sample size was dependent upon the maximum number of participants who could be recruited in a single academic term between January and May 2021. A sensitivity power analysis revealed that the sample was suitably powered to detect a two-tailed correlation of $r = .22$ ($\alpha = .05$, $1 - \beta = .90$); and was in line with recommended minimum sample sizes for PCA ($N > 200$; Kline, 2011).

Study 2

Participants were recruited via social media (i.e., WhatsApp groups and Facebook groups), and the student participant pool in exchange for course credits. An initial sample of 135 participants responded to the survey. Exclusion criteria were the same as Study 1 (i.e., incomplete session and repetitive responding on

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questionnaires), however, in this study, participants were also excluded for completing the study on their phone (as shown by Android or iOS device) due to them being incompatible with the presentation of the task stimuli.

In total 25 participants were excluded based on these criteria, leaving a final sample of 109 participants. The sample consisted of 94 female and 15 male participants. The average age of participants was 31.23 years old ($SD = 13.99$; range = 18–70). The sample size was dependent upon the maximum number of participants who could be recruited across two academic terms between September 2023 and April 2024. A sensitivity power analysis revealed that the final sample was suitably powered to detect a two-tailed correlation of $r = .3$ ($\alpha = .05$, $1 - \beta = .90$).

When analyzing the attention probes presented during the monotonous responding task, an additional 29 participants were excluded for either reporting their attention being focused on situational distractors ($N = 17$; see the Monotonous Responding task section for details); or for giving extreme outlier responses (>3 *SD* from the mean) on the attention probes, which violated normality of the data ($N = 12$; see the Statistical Analysis section). The final sample for this measure was therefore $N = 80$, which had sufficient power to detect $r = .34$ ($\alpha = .05$, $1 - \beta = .90$).

Transparency and Openness

For both studies, all measures collected are reported in the article with no variables omitted. All raw and processed data as well as any analysis scripts used are made available via the Open Science Framework (https://osf.io/wjc45/?view_only=46141f0dc32147509bae08b15e9a9a8f). Neither study was preregistered. All data were collected after review by the ethics committee at the University of Roehampton (ethics code: CB_ERP2_20).

Materials

Interoceptive Attention Questionnaire (Gabriele et al., 2022)

The Interoceptive Attention Questionnaire is a new scale developed to isolate the amount of time that an individual's attention is allocated to bodily signals, independent of their perceived ability to accurately detect these signals. This measure is a 21-item scale, including items such as "Most of the time my attention is focused on whether my heart is beating fast." Responses were along a 5-point scale, ranging from *strongly agree* to *disagree strongly*. Detailed instructions were given to participants prior to the scale to emphasize that responses should be how much attention they paid to their bodily sensations, regardless of how well they thought they could perceive them (see Supplemental Material 1 for full list of items and instructions).

Interoceptive Accuracy Questionnaire (Murphy et al., 2020)

To confirm that the measure of interoceptive attention was dissociated from perceived ability to track and notice interoceptive signals we also measured responses to the Interoceptive Accuracy Questionnaire. This is a recently developed scale designed to isolate the perceived ability to discriminate bodily signals accurately, independent of the amount of time focused on bodily signals. It is a 21-item measure, which includes items such as "I can always accurately perceive when my heart is beating fast." Responses were along a 5-point scale, ranging from *strongly agree* to *disagree strongly*. Detailed instructions were given to participants prior to the scale to emphasize that responses should be how well they believed they could perceive each specific signal, without using external cues, for example, taking pulse for heart rate (see Murphy et al., 2020, for full instructions).

Body Perception Questionnaire (Porges, 1993)

In order to allow comparison to previous samples which explored perceived interoceptive sensitvity/awareness, we measured the Body Perception Questionnaire. This 45-item measure is one of the most widely used measures of self-reported interoceptive awareness and assesses awareness of bodily sensations such as "Muscle tension in my face" or "My mouth being dry." The response was along a 5-point scale ranging from *never* to *always*. This measure is reported in basic correlation analysis for reference, and is included in the complete data available via the Open Science Framework (https://osf.io/wjc45/?view_only=46141f0dc32147509bae08b15e9a9a8f; Brown & Herman, 2025).

Penn State Worry Questionnaire (Meyer et al., 1990)

This questionnaire is a 16-item measure of worry, defined as perseverative thoughts about current or future concerns, and includes items such as "Once I start worrying, I cannot stop." Participants responses are along a 5-point scale, ranging from *not at all typical of me* to *very typical of me*.

Rumination–Reflection Questionnaire (Trapnell & Campbell, 1999)

The Rumination and Reflection Questionnaire is a 24-item measure, divided into two subscales. One subscale is a 12-item measure of rumination, defined as perseverative thinking about oneself and the past, and includes items such as "Sometimes it is hard for me to shut off thoughts about myself." The other subscale is a 12-item measure of self-reflection, defined as abstract or analytical thoughts about oneself, and includes items such as "I often love to look at my life in philosophical ways." Responses are along a 5-point scale ranging from *strongly disagree* to *strongly agree*.

Spontaneous and Deliberate Mind-Wandering Scales (Carriere et al., 2013)

Deliberate and spontaneous mind-wandering were measured through two four-item scales developed by Carriere et al. (2013). Spontaneous mind-wandering was measured with items such as "It feels like I don't have control over when my mind wanders"; while Deliberate mind-wandering was measured with items such as "I find mind-wandering is a good way to cope with boredom."

Responses were along a 7-point scale ranging from either *rarely/not true at all* (1) to *a lot/very true* (7). The items were clarified with the instruction to respond in relation to "how your mind wanders, i.e. when your mind is focused on internal thoughts."

Emotional Awareness and Clarity Scales (Gratz & Roemer, 2004)

The Emotional Awareness and Clarity Scales are subscales taken from the Difficulty in Emotional Regulation Scale. The Emotional Awareness Scale consists of six-items which measures the deliberate attention allocated to identifying current emotions, and includes items such as "When I'm upset I take time to figure out what I'm really feeling." The Emotional Clarity Scale consists of five-items which measure the perceived ability to identify emotions and includes items such as "I have difficulty making sense out of my feelings."

In order to record the adaptive ability of emotional clarity and awareness, rather than deficits in these factors, responses were coded in the opposite direction to the original measure (Gratz & Roemer, 2004). Higher scores therefore reflect more time attending to emotions and ability to accurately recognize emotions most of the time. Responses to both scales were along a 5-point scale ranging from *almost never* (*0%–10%*) to *almost always* (*91%–100%*). In Study 2, Item 1 from the Emotional Awareness Scale was omitted due to an error.

Generalized Anxiety Disorder Scale (Spitzer et al., 2006)

To measure self-reported trait anxiety, participants completed a commonly used anxiety screening tool: the 7-item Generalised Anxiety Disorder Scale. This requires participants to report how often they experienced symptoms of anxiety over the past 2 weeks, including items such as "Feeling nervous, anxious or on edge." Responses were along a 4-point Likert scale ranging from *not at all* to *almost every day*.

Monotonous Responding Task With Three-Dimensional Attention Probes

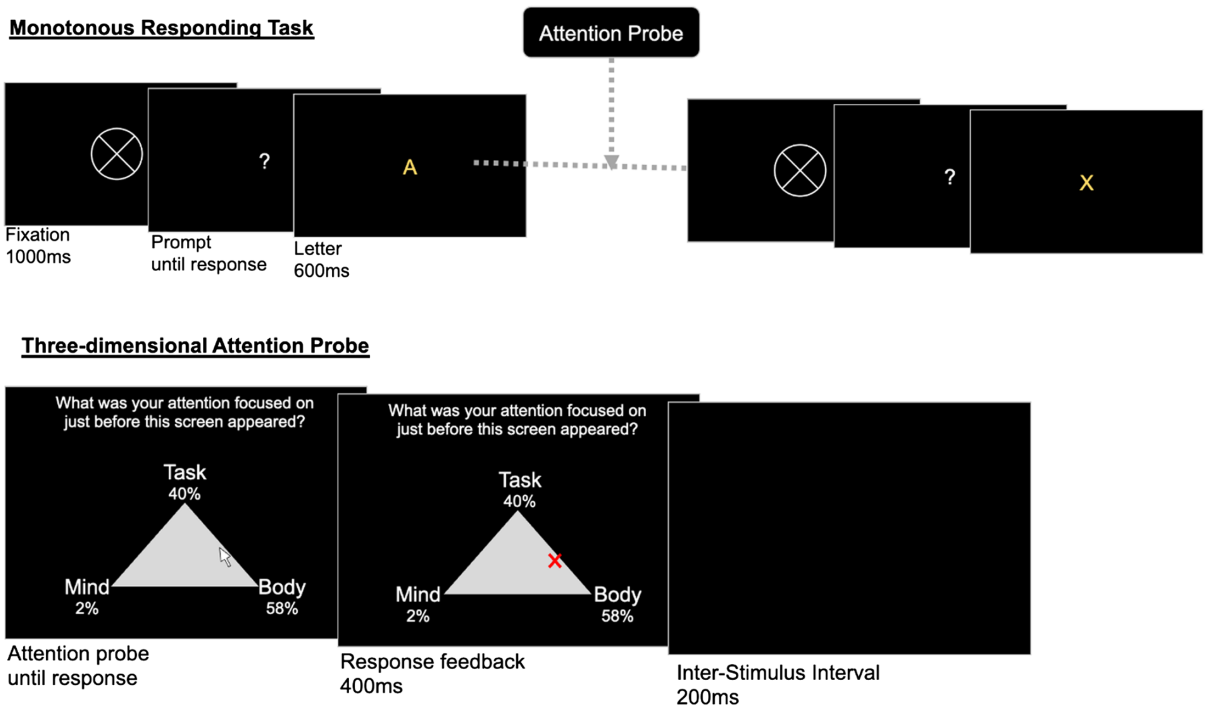
Participants completed a novel monotonous responding task online, which required them to press self-selected buttons on their keyboard at random (see Figure 1). This task was designed to be highly sensitive to competition from internal thoughts and bodily signals as it was simple, uninteresting, and both perceptually and cognitively undemanding; and therefore afforded the capacity to allocate attention to task-irrelevant thoughts/sensations, while also enabling us to remotely measure that they were somewhat engaged behaviorally with the task.

Each trial began with a fixation cross for 1,000 ms, before a white question mark prompt appeared on the screen signaling participants to self-select a button on the keyboard to press. This remained on screen until a response was made, at which point the letter was presented back to the participants in a gold color for 600 ms, before the next trial began. The task consisted of 220 individual prompted response trials. All characters were presented in size 26 Arial font in the center of the screen.

Prior to the task, participants were instructed:

> In this task, you will be asked to press a random letter key on the keyboard every time that the "?" symbol appears on the screen. The order that you press the keys across the task should be random, meaning that each key should have an equal chance of being pressed each time

Figure 1
Illustration of the Novel Monotonous Respond Task and Three-Dimensional Attention Probe



Note. See the online article for the color version of this figure.

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you are prompted, and you should avoid pressing the same keys too many times. You should try to focus on the task as much as possible, though it is likely that your mind will wander to other thoughts and physical sensations while you complete it.

During the completion of the monotonous responding task 10 probe questions were presented at random intervals, though to space them out the minimum interval between two probes was set to 12 monotonous response trials. The probe screen appeared with the question "What was your attention focused on just before this screen appeared?" In the center of the screen was a gray colored triangular response area, the topmost point was labeled with "TASK," the leftmost point "MIND," and right most point "BODY." Below, each label was a percentage value representing the current proximity of the mouse to each point, that is, 100% would reflect maximum closeness to 1 point and maximum distance from the other two points in the triangular space (based on barycentric co-ordinates). Participants were instructed to click on the triangle reflecting where their attention was focused prior to the screen appearing, selecting closest to the label on the triangle which held their attention the most and furthest away from the label which was least attended. Participants were instructed to respond in-between the points if their attention was divided between the Task, Mind, and Body, and to use the percentage values next to the labels to inform their response. After responding, a red "X" appeared for 400 ms at the position participants had selected, and the monotonous response task trials began again after a brief 200 ms pause.

Prior to the task, participants were given the following definitions for the three labels, with "Task" reflecting "all aspects of the button pressing task, for example, possible next letter to choose, timing responses, enjoyment of the task"; "Mind" reflecting: "your internal thoughts which are not about the task, for example, worry/concerns, planning, future/past events"; and "Body" reflecting: "the physical sensations of your body, for example, breathing, itchy skin, hunger, heart rate, muscle tension."

Prior to the task, participants completed a short 8-trial practice block with just the button-pressing task, followed by an 8-trial practice block with the attention probes appearing after 50% of trials.

Situational Distractor Check

In order to isolate only participants who were focused on just the task, their thoughts, or their body sensations, participants were excluded from the attention probe analysis if the repeatedly checked their phone and/or focused their attention on other situational distractors around them during the monotonous responding task. To identify these participants, at the end of the task participants were asked to disclose whether they had used their phone during the button-pressing task. Participants selected the option from a drop-down menu that best reflected their engagement with their mobile phone during the task: "I never engaged with my phone," "I glanced at notifications, but didn't pick it up," "I repeatedly checked my phone, but only briefly," "I repeatedly checked my phone for extended periods of time," and "I spent the majority of the time on my phone." Only those that reported the first 2 options were retained in the data analysis for the attention probes.

The question was prefaced with the statement:

Honestly, during the task, how many times did you engage with your phone, for example, going on social media, checking notifications, mobile games etc. You will not be penalised if you did, but it is important for us to know before we analyse the data. Your honesty is appreciated.

They were also asked whether they had been focusing their attention on other external distractions which were not relevant to the task, their thoughts, or bodily sensations. They selected the option most reflective of their experience from a drop-down menu with the options: "There were no other distractions around me while completing the task," "There were distractions around me while completing the task—I ignored these," "There were distractions around me which I was also focusing my attention on (e.g., watching TV in the background)," and "I focused most of my attention on something else around me, and did not pay much attention to the task or my thoughts/body." Only those that reported the first 2 options were retained in the data analysis for the attention probes.

Procedure

Study 1

Participants accessed the study through Qualtrics software. They first completed a consent procedure, before completing the Spontaneous and Deliberate Mind-Wandering Questionnaire, Penn State Worry Questionnaire, Rumination and Reflection Questionnaire, Interoceptive Attention Questionnaire, Interoceptive Accuracy Questionnaire, and the Emotional Awareness and Clarity subscales of the Difficulty in Emotional Regulation Scale—all in a random order. All participants also completed the Body Perception Questionnaire at the end before being debriefed.

Study 2

Participants access the study through Inquisit software, which they downloaded to their home computer. They first completed a consent procedure, before completing the monotonous responding task, after which they completed the situational distractor check. Participants then completed the Spontaneous and Deliberate Mind-Wandering Questionnaire, Generalized Anxiety Disorder Questionnaire, Penn State Worry Questionnaire, Rumination and Reflection Questionnaire, Interoceptive Attention Scale, and the Emotional Awareness and Clarity subscales of the Emotion Regulation Questionnaire in a random order.

Statistical Analysis

Self-Reported Trait-Level Attention

Across the two studies, the same self-reported trait-level attention to thoughts were measured, therefore the following analyses for self-report measures were repeated in both samples. The results across the two studies are therefore merged. All analyses were conducted using Jeffreys's Amazing Statistics Program software (Jeffreys's Amazing Statistics Program Team, 2023).

To identify the core components of internally focused attention to thoughts and reduce the number of variables to their core components, PCA was conducted with the standardized mean scores of the Spontaneous and Deliberate Mind-Wandering Scales, Penn State Worry Scale, Rumination and Reflection Scales. Components

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were extracted through the combination of scree plot and parallel analysis, with the 95th percentile of all Eigenvalues from the simulated data overlaid on the screen plot to identify the breakpoint (see Supplemental Material 2). An orthogonally rotated solution was calculated with Varimax rotation to extract discrete components. Coefficients loading below .3 on either component were suppressed; additionally, scales which resulted in cross-loadings with a difference less than .3 between components were also suppressed (based on Costello & Osborne, 2005). Composite scores were then calculated by standardizing the individual scales (i.e., *Z* scores) identified within the PCA, and averaging them together to create a composite standardized score (see C. J. R. Smithson et al., 2024, for discussion of this approach). These composite scores were then used to assess the key research question in the current investigation by assessing their relationship with interoceptive attention.

To assess whether different internal thought components were more predictive of interoceptive attention, regression analyses were conducted to isolate the unique predictive contribution of each of the thought composite scores derived from the PCA. Based on evidence that age and gender are notable predictors of interoceptive attention and could bias the analysis (Gonçalves & Byrne, 2013; Grabauskaitė et al., 2017; Ma-Kellams et al., 2024; Naveteur et al., 2015), these were also entered into the regression model as covariates. For purely statistical reasons, only male and female participants were analyzed, resulting in the inclusion of 220 participants in this analysis. Repetition of these analyses without covariates did not change the level of significance or interpretation. The relationship between these covariates to the current data and other outcome and predictor variables was also assessed in separate analyses.

Further, to control for other potential confounding variables and to assess the unique contribution of internally focused attention, follow-up regression analyses were conducted with self-reported interoceptive accuracy added as a covariate in Study 1, and trait anxiety in Study 2. Finally, to assess whether the measures of internal thoughts and interoceptive attention showed similar relationships with emotional awareness and clarity, zero order correlations were conducted with these variables.

Three-Dimensional Attention Probes

Compositional Data Analysis. The three-dimensional attention probe structure results in inherently compositional (i.e., ipsative) data, where the finite shared variance always sums to 100, regardless of participants' priorities. This tightly intercorrelated set of variables poses challenges for conventional statistical analyses. We therefore employed a compositional data analysis approach (Aitchison, 1982; Greenacre et al., 2021). The approach applies a centered-log ratio (CLR) transformation to the data, whereby each compositional priority score is divided by the central geometric mean, thus providing the distance of each score from the central "barycentric" point within the rating structure (i.e., reflecting no biased priority). This center score is then log-transformed to map the originally ipsative data to an unrestricted data space suitable for further analysis (M. Smithson & Broomell, 2024). All compositional data analyses were conducted in Python, see Open Science Framework for available code (https://osf.io/wjc45/?view_only=46141f0dc32147509bae08b15e9a9a8f).

In cases where a centered priority value was zero and could not be log-transformed, zero was replaced by .0065, which reflects the minimum detectable score (.01) multiplied by .65 (based on recommended guidelines; Lubbe et al., 2021; Martín-Fernández et al., 2003). After transformation, histograms were visually inspected to identify extreme responses which violated normality assumptions. A default cutoff of 3 *SD* was selected which removed these outliers, resulting in normally distributed data. We note that reanalysis of the data with outliers retained did not alter the overall interpretation of the findings.

After data were CLR transformed, the reliability of the responses for body, mind, and task priority probe values was assessed across the 10 time points, using average fixed rater intraclass correlation coefficients. This was conducted using the *intraclass_corr* function from the *Pingouin* Python library (Vallat, 2018).

To assess the relative priority scores across the duration of the task, a restricted maximum likelihood linear-mixed model analysis was conducted using the *mixedlm* function from the *statsmodels* Python library (Seabold & Perktold, 2010). The CLR-transformed probe priority values were entered as the outcome variable, with probe components (Task, Mind, Body) and time point (10 time points) entered as interacting factors. Participant ID was entered as a random effects factor to capture the within-subjects structure of the data.

Finally, to test the strength of the competitive relationships between the task, mind, and body probe priority values, and whether these varied over time, three separate linear-mixed model analyses for each of the three pairwise correlations were conducted (i.e., Body × Mind, Body × Task, Mind × Task). Within these analyses, time point was entered as a predictive factor, and Participant ID was entered as a random effects factor to retain the within-participants design. Follow-up individual correlation coefficients were calculated for each time point for each of the three linear-mixed model analyses, to assess the strength of the relationship at each time point. These were Bonferroni corrected for the 30 follow-up correlations (adjusted $\alpha < .002$).

Results

Self-Reported Trait Attention to Thoughts Scales

Assessment of all self-report measure revealed good internal consistency, with all scales showing good to excellent levels of reliability ($\alpha > .80$; see Table 1).

Data Reduction of Attention to Thoughts Scales

The PCA revealed two components explaining 72.91% of the variance (Table 2), with the Kaiser–Meyer–Olkin measure of sampling adequacy equaling .67, above an acceptable .6 threshold (Kaiser, 1974). Bartlett's test of sphericity was also significant, $\chi^2(10) = 344.68$, $p < .001$. Of the two components, the first reflected automatic internally focused attention to negative thoughts, explaining 50.30% of the variance; and was composed of worry and rumination (Spearman–Brown correlation: $r = .73$, $p < .001$). The second component reflected more voluntary and controlled internally focused attention to thoughts, explaining 22.60% of the variance; and consisted of deliberate mind-wandering and reflection (Spearman–Brown correlation: $r = .37$, $p < .001$). Spontaneous mind-wandering loaded partially on both components and was therefore excluded.[1]

[1] Repeating the PCA analysis on the data from the Study 2 sample revealed the identical components (see Supplemental Material 2).

Table 1
Descriptive and Reliability Statistics for All Self-Report Measures

Self-report measure	Study 1 (N = 222) Mean total score	*SD*	Cronbach's α	Study 2 (N = 109) Mean total score	*SD*	Cronbach's α
Interoceptive attention	61.75	14.51	.93	50.92	13.95	.91
Interoceptive accuracy	81.5	12.18	.92			
Body Perception Questionnaire	129.37	39.45	.97			
Rumination	42.86	9.03	.90	45.03	8.65	.90
Reflection	41.44	6.78	.80	41.21	8.60	.90
Worry	55.35	13.96	.92	55.66	14.1	.95
Spontaneous mind-wandering	17.89	5.38	.81	20.11	5.35	.82
Deliberate mind-wandering	18.09	5.52	.79	19.25	6.25	.87
Emotional awareness	21.5	4.97	.82	17.56	4.92	.89
Emotional clarity	17.54	3.74	.73	17.93	4.51	.85
Generalized anxiety				9.29	5.62	.90

Correlations

The composite scores, calculated from the extracted components, were positively related in Study 1, but were nonsignificantly related within Study 2 (see Tables 3 and 4, for all correlations; see Supplemental Material 3, for full correlations between individual scales).

To determine whether controlled thoughts and negative automatic thoughts are independently related to interoceptive attention, these scores were entered as predictors within a linear regression analysis, with age and gender included as covariates. Independent correlation and t test analyses of the covariates' effect on other variables revealed that female participants reported significantly higher levels of interoceptive attention, and controlled and negative automatic thought scores; age correlations revealed that older participants reported lower levels of controlled and negative automatic thought scores (see Table 1).

The overall regression analysis revealed that in Study 1 only the negative automatic thoughts score uniquely correlated with interoceptive attention, $\beta = .27$, $p < .001$, and the controlled thoughts score was nonsignificant, $\beta = .12$, $p = .069$. Within Study 2, the same pattern emerged with only the negative automatic thoughts score reaching significance, $\beta = .31$, $p = .001$, and the controlled thoughts score again failing to reach significance, $\beta = .02$, $p = .839$.

Table 2
Components Derived From a Principal Components Analysis

Self-report measure	Component: Negative automatic thought	Controlled thought
Worry	0.912	
Rumination	0.883	
Spontaneous mind-wandering	0.663	0.461
Reflection		0.817
Deliberate mind-wandering		0.793
Eigenvalues	2.52	1.13

Note. Scale scores loading uniquely on each component are highlighted in gray. Only loadings with an absolute value greater than .2 are displayed for visibility.

Role of Perceived Interoceptive Accuracy

To confirm that the observed relationships between negative automatic thoughts and interoceptive attention were due to reported time attending bodily signals, and not the perceived ability to do so accurately, we repeated the regression analysis but included perceived interoceptive accuracy as an additional covariate which was measured within Study 1. This revealed that, within Study 1, the relationship between interoceptive attention and negative automatic thoughts remained significant, $\beta = .28$, $p < .001$. Controlled attention remained a nonsignificant predictor of interoceptive attention, $\beta = .11$, $p = .102$. Within the model, perceived interoceptive accuracy was also a nonsignificant predictor, $\beta = .07$, $p = .301$. Thus, it appears that the previous results reflected perceived time attending to bodily signals, rather than perceived accuracy.

Role of Trait Anxiety

Within the Study 2 sample, trait anxiety was found to positively correlate with both interoceptive attention and negative automatic thoughts. To further explore whether this result was consistent with trait anxiety being an underlying common factor for both forms of attention, a partial correlation was conducted between the negative automatic thoughts score and interoceptive attention, while controlling for the trait anxiety score. Within this analysis the relationship became weaker and nonsignificant, $r_{partial} = .12$, $p = .206$. It should be noted, however, that some of the items within the 7-item Generalised Anxiety Disorder Scale directly refer to worry, as it is a core symptom of anxiety, which could be causing this pattern of results. Repeating the analysis again with these items removed (i.e., Items 2 and 3) also produced a similar result, with a weaker and nonsignificant relationship between negative automatic thoughts and interoceptive attention, $r_{partial} = .17$, $p = .079$.

Relationship With Emotional Awareness and Clarity

Within both Studies 1 and 2, negative automatic thoughts and interoceptive attention both showed a similar relationship with emotional clarity and awareness: both inversely correlated with the clarity inidentifying emotions, and both were nonsignificantly related to emotional awareness—with these results replicated in both

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Table 3
Study 1 (N = 222) Zero Order Correlations Between Variables and Demographic Variables

Self-report measure	1	2	3	4	5	6	7	Age	Gender Cohen's *d* (female vs. male; *N* = 220)
1. Negative automatic thoughts score	—							−.32***	.87***
2. Controlled thoughts score	.26***	—						−.27***	.40*
3. Interoceptive attention	.30***	.19**	—					−.08	.40*
4. Interoceptive accuracy	−.03	.114	.07	—				.05	.26
5. Body Perception Questionnaire	.11	.18**	.14*	.16*	—			−.10	.09
6. Emotional awareness	−.06	.30***	.02	.30***	.19**	—		.02	.18
7. Emotional clarity	−.31***	.10	−.20**	.27***	.01	.49***	—	.20**	−.25

Note. Positive gender difference effects sizes reflect higher female values.
* $p < .05$. ** $p < .01$. *** $p < .001$.

Studies (see Tables 3 and 4). The controlled thoughts score on the other hand was nonsignificantly correlated with emotional clarity in both Studies 1 and 2, but was positively correlated with emotional awareness in Study 1, but not Study 2. Thus, interoceptive attention and negative automatic thoughts showed similar relationships, while controlled thoughts showed a divergent relationship consistent with another mechanism.

State-Level Attention

The intraclass correlation coefficients analyses revealed good levels of internal consistency across the time points for each of the three probe scores (see Table 5; Koo & Li, 2016). The linear-mixed model assessing the CLR-transformed priority values confirmed that all three priority values were significantly different from one another. Attention to the task was rated higher than attention to thoughts (i.e., Mind), $M_{\text{diff}} = 1.36$, $p < .001$, 95% CI [1.17, 1.55], and bodily sensation (i.e., Body), $M_{\text{diff}} = 3.37$, $p < .001$, 95% CI [3.18, 3.56]; and attention to thoughts was prioritized more versus bodily sensation, $M_{\text{diff}} = 2.01$, $p < .001$, 95% CI [1.82, 2.20].

Interestingly, time also interacted with each of these contrasts, with attention to the task showing a linear decrease over time, $B = -.15$, $SE = .02$, $p < .001$, 95% CI [−0.19, −0.11]; while attention to thoughts significantly increased, $B = .25$, $SE = .03$, $p < .001$, 95% CI [0.20, 0.31], along with attention to bodily sensations, $B = .20$, $SE = .03$, $p < .001$, 95% CI [0.15, 0.26]. Thus, this analysis indicates that though the priority of each of the different probe components was significantly different and this remained consistent (see Figure 2), there was variation in priority over time, with attention becoming less task focused.

Competitive Relationships Over Time

The overall cumulative correlation between the CLR-transformed probe reports revealed that attention to bodily signals was inversely correlated with both the attention to the task, $r = -.63$, $p < .001$, as well as attention to internal thoughts, $r = -.46$, $p < .001$. Similarly, attention to thoughts was also inversely correlated with attention to the task, $r = -.40$, $p < .001$, thus suggesting a competitive relationship at the state-level between attention to the task, bodily signals, and internal thoughts.

When assessing the variation in the strength of these relationships, it was found that there was a small but significant increase in the strength of the negative relationship between attention to body sensations versus the task, $r_{\text{change}} = -.01$, $SE = .001$, $p < .001$, 95% CI [−0.012, −0.010], with this relationship remaining strong over time ($r > \pm .64$; Table 6).

Conversely, the relationship between attention to the task and thoughts became significantly weaker, $r_{\text{change}} = .09$, $SE = .001$, $p < .001$, 95% CI [0.090, 0.096], and by the midpoint of the task there was no significant competition between these components of the attention probe, with correlations becoming small and nonsignificant ($r < \pm .18$).

Interestingly for our hypothesis that internal attention to our thoughts comes at the cost of attention to our bodily signals, we found over time there was increased competition between these components of the probe as shown by increasingly negative correlations, $r_{\text{change}} = -.07$, $SE = .001$, $p < .001$, 95% CI [−0.074,

Table 4
Study 2

Self-report measure	1	2	3	4	5	6	Age	Gender Cohen's *d* (female vs. male)
1. Negative automatic thoughts score	—						−.13	.40
2. Controlled thoughts score	.11	—					−.32***	.52
3. Interoceptive attention	.31***	.08	—				−.16	−.08
4. General Anxiety	.71***	.18	.33***	—			−.04	.16
5. Emotional awareness	−.15	.06	−.17	−.20*	—		.04	−.17
6. Emotional clarity	−.47***	−.03	−.23*	−.51***	.57***	—	.18	.29

Note. $N = 109$. Zero order correlations between variables and demographic variables. Positive gender difference effects sizes reflect higher female values.
* $p < .05$. *** $p < .001$.

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Table 5
Study 2 Descriptive Statistics and Intraclass Correlations of the Attention Probes

Probe component	Average raw response score (*SD*)	Average CLR-transformed response score (*SD*)	ICC	95% CI *LB*	95% CI *UB*
Task	70.13% (26.79)	2.02 (1.82)	.84	.78	.89
Mind	19.20% (21.16)	−0.39 (1.58)	.81	.74	.87
Body	10.70% (15.70)	−1.63 (1.87)	.70	.60	.79

Note. Study 2 descriptive statistics and ICC of the attention probes (n = 10) during the monotonous responding task, after additional exclusions for outliers, and self-reported distracted participants (final N = 80). ICC = intraclass correlation coefficients; CLR = centred-log ratio; CI = confidence interval; *LB* = lower bound; *UB* = upper bound.

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−0.069]. Though initially these probe scores were nonsignificantly and weakly correlated ($r > \pm .11$), by the midpoint of the task a strong and significant negative relationship emerged ($r > \pm .59$).

The overall results therefore indicate that attention to our bodily signals competes against other factors, but that competition varies over time. In the current data, the initial competition was between attention to task versus internal thoughts and bodily signals; however, as the time went on and attention was less focused on the task, attention to bodily signals and internal thoughts began to compete.

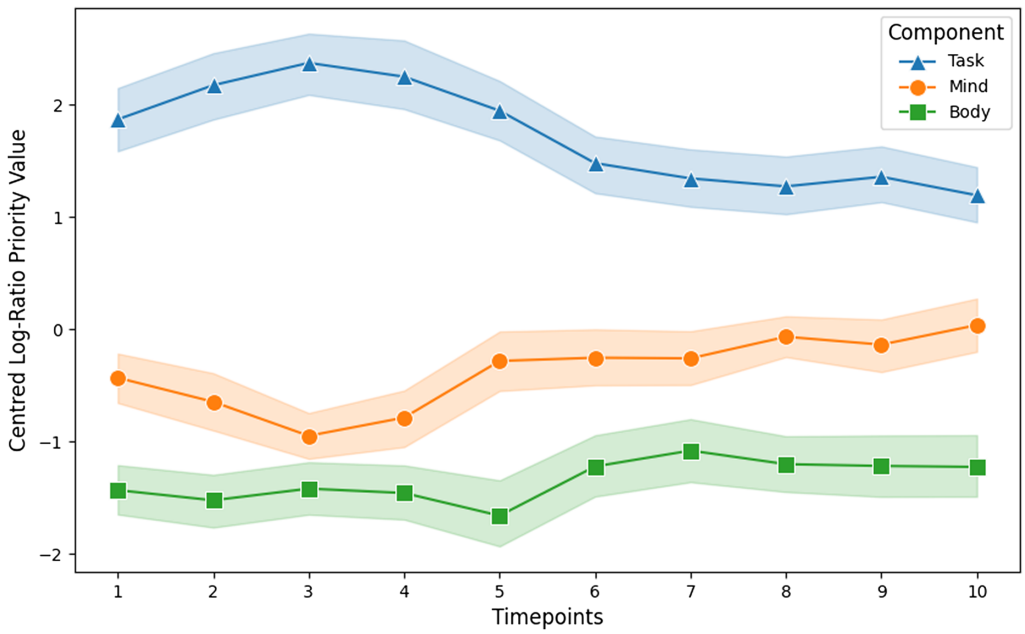
Exploratory Correlations

To explore potential correlations beyond the intended internal correlations of the probes, the overall CLR-transformed body probe priority score was correlated with self-reported interoceptive attention, negative automatic thoughts scores, and controlled thoughts scores (with composite Z score variables recalculated for $N = 80$). The results revealed that the only significant correlation with the average Body probe score was between self-reported interoceptive attention, $r = .26$, $p = .020$. While controlled and negative automatic thoughts scores were positively correlated, these failed to reach significance, controlled: $r = .13$, $p = .262$, and negative automatic thoughts: $r = .12$, $p = .284$.

Discussion

In the current investigation we explored whether attention to interoceptive signals and attention to internal thoughts could reflect a common internally focused attention, or whether they compete for finite attentional resources. We revealed divergent relationships whereby trait-level self-reported attention to internal thoughts, especially negative automatic thoughts, positively correlated with self-reported interoceptive attention; however, when assessed at the state-level using a novel three-dimensional attention probe measure, we found a competitive relationship.

Figure 2
Plot Depicting the Center Log-Ratio Transformed Attentional Priority Divided Between Task, Mind, and Body Across 10 Time Points



Note. The error band reflects 95% confidence intervals. See the online article for the color version of this figure.

Table 6
Pairwise Correlations Between Centered Log-Ratio Priority Values Across 10 Time Points

Probe component relationship	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Mind × Task	−.60***	−.58***	−.43**	−.58***	−.18	−.17	.13	.002	.17	.04
Body × Task	−.73***	−.70***	−.68***	−.66***	−.64***	−.69***	−.76***	−.80***	−.79***	−.75***
Body × Mind	−.11	−.17	−.37**	−.24	−.64***	−.59***	−.74***	−.60***	−.74***	−.69***

Note. $N = 80$. Darker colours signify stronger relationships, with blue signifying negative correlations, and red signifying positive correlations. T1–T10 = time point 1 to time point 10. Bonferroni corrected for 30 correlations. See the online article for the color version of this table.
** $p < .0001$. *** $p < .00001$.

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State-Level Competitive Relationships

The current findings from the state-level three-dimensional attention probes are consistent with previous research findings, which reveal competitive relationships between external task stimuli, interoception, and mind-wandering. For instance, evidence showing that awareness of sensory experiences is reduced when engaging in a concurrent cognitive task (Qu et al., 2021; Salgues, Plancher, & Michael, 2021). Similarly, when participants are instructed to deliberately attend to bodily sensations, there is a decrease in the frequency of mind-wandering (Levinson et al., 2014; Weng et al., 2020). Within these previous investigations, however, participants are externally instructed to deliberately prioritize one stream of information over the other, and when a concurrent task is given it is often cognitively demanding, and therefore occupies the limited capacity required to focus on more than one stream of information. It is therefore difficult to determine whether this competition is specific to the experimental task or whether competition between interoceptive attention and other streams of information emerges in contexts without explicit instruction or cognitive demands.

The current investigation, therefore, reveals for the first time that even without instructions to focus on either internal thoughts or bodily sensations, attention to these distinct streams of sensory/cognitive information compete. In line with the idea that there are distinct internal and external attentional (or interoceptive and exteroceptive) mechanisms (Chun et al., 2011; Farb et al., 2013), the decline in attention to the task resulted in increased attention to both internal thoughts and bodily sensation. Analysis of the correlations between priority scores across the task, however, reveals that this does not reflect a binary internal versus external mechanism. Specifically, though the increase in attention to bodily sensations and thoughts initially both occurred because of a reduction in attention to the task, this did not remain consistent. Increasingly, attention to internal thoughts inversely correlated with attention to body signals, and less with the task. The results therefore demonstrate that rather than attention being drawn inward to both "internal" streams of information equally when focus on the task declined, attention was increasingly allocated to either the bodily sensations *or* internal thoughts.

This competitive relationship was found even though the simple monotonous responding task was designed to maximize attentional capacity. In more demanding tasks which require greater engagement with external stimuli there may be even less spare attentional capacity, further exaggerating the competitive relationship found here. Beyond task characteristics, other factors also likely influence the allocation of state-level attention between different internal streams of sensory/cognitive information. One such factor could be the negative emotional state of the individual or task content, as it has been shown that we often shift our attention to internal thoughts when experiencing negative emotions (Smallwood et al., 2009); however, it has also been found that interoceptive attention is also increased under negative mood (Poerio et al., 2024). Therefore, at the state-level it is not clear yet which would be prioritized when an individual was experiencing negative mood, or which moderating individual factors would play a role in this balance. Future research should therefore explore how variation in contextual demands and internal states influence the allocation of attention to bodily signals in competition with other streams of information.

Broadly, fundamental established theories of selective attention (e.g., Anderson, 2016; Lavie, 1995) suggest that when attention toward a stimulus dimension is incentivized or there is limited capacity to focus on multiple stimuli, this results in the prioritization of a limited set of stimuli at the expense of others. Interoceptive attention does not seem to be an exception to this, however, this must still be confirmed. For instance, there is a possibility that the modality of internal thought may be a factor (e.g., verbal or visual internal thought), as spontaneous physical sensations have been found to positively correlate with the frequency of inner speech (Salgues, Plancher, et al., 2021).

Trait-Level Convergent Relationships

In the current investigation, the trait-level measures of thought were derived from the PCA of multiple measures of internal thought. Interestingly, this produced two separate internal thought components, one reflecting negative automatic thoughts (i.e., worry and rumination) and another reflecting controlled thoughts (i.e., deliberate mind-wandering and reflection), consistent with other dual-mechanism models of attention and thought (e.g., Seli et al., 2015).

Though the state-level findings reveal that in a single moment internal thoughts may compete with attention to our physical sensations, the trait-level self-reported attention suggests that, over time, general attention to these two streams of information may be correlated. More specifically, the results showed that individuals who report high levels of interoceptive attention also report higher levels of negative automatic thoughts in both samples, independent of controlled thought levels. This relationship also remained significant when controlling for perceived interoceptive accuracy, indicating that the relationship was with the recalled time attending to interoceptive signals, rather than the perceived ability to discriminate these signals.

Controlled thoughts on the other hand exhibited a weaker positive relationship and only reach conventional levels of significance in the larger sample of Study 1, when not controlling for negative

automatic thoughts. Additionally, it also exhibited weaker and more nonsignificant correlations with all other variables in Study 2, relative to Study 1. This lack of replication is likely due to the lower statistical power in the smaller Study 2 sample, as the required sample to detect the correlation between controlled thought and interoceptive attention found in Study 1 ($r = .19$) would be $N = 234$ ($\alpha = .05$; $1 - \beta = .90$). While we cannot confidently conclude that controlled thoughts are linked to interoceptive attention, it is likely a smaller effect if it were a predictor.

The current pattern of results appears to suggest that interoceptive attention may be more strongly linked with negative automatic thoughts, rather than a disposition to focus attention inward to all current thoughts. This appears to contradict other models of interoception which suggest that nonemotional controlled mechanisms are also involved in perceived interoceptive awareness/sensibility, as well as more automatic anxiety related cognitions (Baiano et al., 2021; Mehling, 2016; Trevisan et al., 2021). With this being reflected in other self-reported measures of perceived interoceptive awareness/sensibility, for instance, the Multidimensional Assessment of Interoceptive Awareness (Mehling et al., 2012, 2018) includes items reflective of adaptive mindful perception.

This could be due to differences between perceived interoceptive attention and perceived interoceptive accuracy (Gabriele et al., 2022; Garfinkel et al., 2015); though an alternative reason could be due to the measure of interoceptive attention utilized in our investigation. Specifically, we used the recently developed Interoceptive Attention Scale (Gabriele et al., 2022), which records the perceived time allocated to different interoceptive signals, for example, "most of my time is focused on. …" Given the perseverate and repetitive nature of negative automatic thoughts (Davey & Meeten, 2016), this form of attention when focused on interoceptive signals may occupy more time. Conversely, a more controlled mechanism of interoceptive attention would likely occupy less time, as the goal-directed process would be terminated once completed (e.g., stopping meditation). Thus, the measure may be biased toward perseverative negative automatic thinking about interoceptive signals. A measure that separates out perceived interoceptive attention into different controlled and automatic mechanisms may isolate different effects.

It should be noted that the positive relationship between trait interoceptive attention and attention to internal thoughts was cross-sectional and could reflect a number of different mechanisms: It could be indicative of (a) a single common factor, (b) a causal relationship between these distinct factors, or (c) an underlying factor contributing to both mechanisms, causing a relationship to emerge.

Based on the current evidence, the initial possibility that interoceptive attention is simply a subtype of internally focused attention to streams of thought seems unlikely. Though somatic sensations have been reported as the contents of mind-wandering in some samples (Diaz et al., 2013, 2014), it is rarely the main focus, and the contents of internal thought are often related to ongoing external concerns or goals (Klinger, 2013). Indeed, in the current data (Table 4) attention to bodily signals was the lowest priority probe component, relative to thoughts and the external task. Instead, physical sensations are likely attended to when individuals are deliberately vigilant for them or when particularly salient (Ginzburg et al., 2014). Additionally, the relationships found between interoceptive attention and attention to negative automatic thoughts were only moderate (i.e., $r = .28$–$.31$), inconsistent with two facets of a single construct.

Alternatively, it could be that interoceptive attention and attention to internal thoughts are distinct processes which causally influence each other, as attention to physical signals could trigger subsequent concerns, or thoughts about a specific concern could cause a shift of attention to a related physical signal. For instance, attending to bodily signals of hunger could trigger subsequent thoughts about food, or thinking about a recent injury could cause attention to be focused on muscle soreness or pain. Consistent with this causal effect, experimental findings have revealed that induced hunger and sexual arousal increases mind-wandering about these states, consistent with this mechanism (Rummel & Nied, 2017). Further, evidence has found that after being instructed to think about planning an unpleasant task, individuals who were prone to rumination reported an increase in awareness of physical sensations, relative to those not under stress—consistent with negative internal thoughts increasing attention to physical sensations (Schlinkert et al., 2020).

The final possibility is that the observed relationship between interoceptive attention and internal attention could be due to an underlying trait which independently influences both factors. One candidate for this variable is trait anxiety, as controlling for this variable resulted in the positive correlation between interoceptive attention and negative automatic thoughts becoming nonsignificant, consistent with this common underlying factor accounting for their relationship. Indeed, trait anxiety and depression have been linked to both negative automatic thinking and to greater vigilance to bodily sensations (Olatunji et al., 2007; Zvolensky & Forsyth, 2002).

This result will require further investigation though, as the potential underlying factor may not be trait anxiety, which itself could be confounded. Instead, it could be that more specific conditions such as hypochondriasis or somatization disorder, which are strongly linked to anxiety but are characterized by more specific body focused fears, could account for the relationship (Löwe et al., 2022). Or alternatively, the confound could be broad personality traits linked to trait anxiety, such as general negative affect/emotionality or neuroticism. These constitute a dispositional tendency to experience more frequent and intense negative emotions in response to stress, and while not always experienced as anxious symptoms they are linked to a greater risk of anxiety and psychopathology in general (Hankin et al., 2017; Tackett et al., 2013).

In support of this, a recent experience sampling study conducted by Poerio et al. (2024) probed attention and emotional states over a week-long period, and found that trait anxiety was uncorrelated with the frequency that attention was allocated to interoceptive signals, but negative mood was positively correlated with this measure. Further, Denovan et al. (2019) found that awareness of somatic complaints (e.g., tightness of chest, dizziness) positively correlated with rumination and worry, though structural equation modeling suggested that these factors were mediating the distal effect of neuroticism.

Future work is therefore required to identify the exact cognitive-emotional mechanism or trait which results in the prioritization of interoceptive signals. As it could reflect variation at a range of levels, rather than just anxiety.

Relationship With Emotional Awareness and Clarity

As well as assessing the relationship between interoceptive attention and attention to thoughts, we also explored their relationship with the perception and awareness of emotions (i.e., emotional clarity

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and awareness), which are important for regulating emotions and may be influenced by these forms of attention. The results revealed that higher levels of both interoceptive attention and negative automatic thoughts correlated with lower levels of emotional clarity in both samples. This similar relationship further supports the idea that both interoceptive attention and attention to negative automatic thoughts reflect a common underlying anxiety-linked factor. Emotional awareness on the other hand did not correlate with either of these measures of self-reported attention, but did positively correlate with controlled thoughts in Study 1, consistent with a more mindful attention required for emotional awareness, though this did not reach significance in Study 2.

In regard to the negative relationship between the measures of attention and emotional clarity, this could be reflective of a direct relationship, as evidence has shown that perseverative negative automatic thoughts, such as worry, occupy attentional capacity (Stefanopoulou et al., 2014), and reduce the efficiency with which individuals can allocate their attention to an ongoing task, impairing task performance (Eysenck et al., 2007). Even when focusing on a single task, perseverative thinking can interfere with attentional mechanisms required to flexibly track and update information in working memory (Gustavson & Miyake, 2016; Gustavson et al., 2020). If they were allocated in a perseverative and negative affective way, both negative automatic thoughts and interoceptive attention could impair emotional clarity.

Alternatively, the similar relationship between the two forms of internally focused attention and emotional clarity could be due, again, to the underlying common factor of anxiety. The current data are not designed however to disentangle the precise mechanisms, and future work will therefore need to explore the direct and indirect relationships involving internally focused attention to bodily signals and thoughts, in relation to emotional processing and regulation.

Limitations and Future Directions

Within the current investigation, the interpretations are based on a cross-sectional correlational design, making it difficult to assess the direction of causation. As such we have proposed several possible mechanisms that could explain the current pattern of results. Future research should test these mechanisms, either with a longitudinal or experimental design in order to infer causation.

We also note that while suited to identifying the balance of attention between all components of the attention probe, the novel three-dimensional attention probe measure may not be ideally suited to exploring individual variation with other measures. Despite the specialized compositional analysis conducted on the data from Study 2, which can decompose the intercorrelated variables (Aitchison, 1982; M. Smithson & Broomell, 2024), the ipsative data from this task may still obscure correlations between interoceptive attention and other variables not incorporated in the attention probe task, due to the influence of other probe responses diluting any pure measure of the specific variable (Johnson et al., 1988). Future work intending to assess individual differences in interoceptive attention using attention probes should use another measure such as experience sampling with Likert scales. Indeed, evidence from Poerio et al. (2024) using this approach over a week-long period found that interoceptive attention is positively correlated with negative mood and fatigue, consistent with our current findings.

Constraints on Generality

Within the current investigation we prioritized using opportunity sampling to reach an adequate sample size meaning that targeted recruitment of non-White, educated, industrialized, rich, democratic (Henrich et al., 2010) participants was not possible given our current resources. We do note that while ethnicity was not formally recorded, the current samples were likely ethnically diverse, as in the U.K.-based student population we recruited from non-White students made up 64.3% of the total population across the 2021–2022 period, and 78.8% of the population in the 2023–2024 period when recruitment was conducted (Roehampton University, 2025). Future research must however formally identify whether the current pattern of results generalizes to non-White, educated, industrialized, rich, democratic cultures, especially as there is reported variation in somatic awareness across cultures (Ma-Kellams, 2014).

Another point for future consideration is the generalization of the self-reported factors to behavioral measures of interoceptive awareness and accuracy. For instance, testing whether both negative automatic thoughts and/or controlled thoughts predict less accurate interoceptive judgments or lower awareness of these judgments, as they compete for the finite attention to perform an interoceptive task (e.g., heart-beat counting task; Murphy et al., 2020).

We note that the use of self-report measures of trait attention could potentially be biased by demand characteristics or recall bias. Despite the subjective nature of these measures, however, evidence does suggest that individuals are often able to accurately self-assess their attentional ability and focus of their attention. Indeed, self-reported trait interoceptive attention has been found to predict the frequency of body focused thoughts when probed using experience sampling methods (Poerio et al., 2024), similarly self-reported trait measures of automatic and controlled thought correlate with the frequency of these specific thoughts when probed during the performance of a cognitive task (Seli, Risko, & Smilek, 2016). Though the self-report measures of attention used here are likely valid, replication of the current finding in laboratory tasks would increase the convergent evidence for our findings.

Conclusion

We reveal for the first time that state-level allocation of attention to internal streams of thought and interoceptive attention are inversely correlated and compete for finite attentional capacity; however, at the self-reported trait-level they exhibit a positive correlation. This trait-level relationship suggests that individuals who attend to their bodily signals also show a preoccupation with negative automatic thoughts comprising rumination and worry. While future research is needed to clarify the nature of this relationship, it is likely that it reflects an underlying anxiety-linked biasing of attention toward these streams of internally focused information.

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